space figures like a circle on a flat table, do not hold from the perspective of an accelerated observer. Of course, we have discussed only one particular kind of accelerated motion, but Einstein showed that a similar result—the warping of space—holds in all instances of accelerated motion.

In fact, accelerated motion not only results in a warping of space, it also results in an analogous warping of time. (Historically, Einstein first focused on the warping of time and subsequently realized the importance of the warping of space.6) On one level, it should not be too surprising that time is also affected, since we have already seen in Chapter 2 that special relativity articulates a union between space and time. This merger was summarized by the poetic words of Minkowski, who during a lecture on special relativity in 1908 said, “Henceforward space on its own and time on its own will decline into mere shadows, and only a kind of union between the two will preserve its independence.”7 In more down-to-earth but similarly imprecise language, by knitting space and time together into the unified structure of spacetime, special relativity declares, “What’s true for space is true for time.” But this raises a question: Whereas we can picture warped space by its having a curved shape, what do we really mean by warped time?

To get a feel for the answer, let’s once again impose upon Slim and Jim on the Tornado ride and ask them to carry out the following experiment. Slim will stand with his back against the ride, at the far end of one of the ride’s radial struts, while Jim will slowly crawl toward him along the strut, starting from the ride’s center. Every few feet, Jim will stop his crawling and the two brothers are to compare the readings on their watches. What will they find? From our stationary, bird’s-eye perspective, we can again predict the answer: Their watches will not agree. We come to this conclusion because we realize that Slim and Jim are travelling at different speeds—on the Tornado ride, the farther out along a radial strut you are, the farther you must travel to complete one rotation, and therefore the faster you must go. But from special relativity, the faster you go, the slower your watch ticks, and hence we realize that Slim’s watch will tick more slowly than Jim’s. Furthermore, Slim and Jim will find that, as Jim gets closer to Slim, the ticking rate of Jim’s watch will slow down, approaching that